Faculty Allotment Management System

Stage 1 Specification (draft for approval)

Status	Draft. Not built. Awaiting confirmation of the assumptions in section 8.
Date	13 October 2026
Version	0.1
Scope	Single college, one admin user, scheduling, approval, and printed room sheets

1. What it does

One admin imports teachers and rooms, sets the exam days and sessions, and generates an allotment. The system checks every rule, the admin approves the allotment, and room sheets are printed for each exam session. Absences and substitutes are handled in Stage 2.

2. Hard rules

These rules are never violated. Each one is enforced in three places: the scheduling engine, the server (for every manual edit and substitute), and the database.

ID	Rule	What it means
H1	Room coverage	Every active room gets exactly the configured number of teachers in each session (default 1).
H2	One session per teacher per day	A teacher has at most one duty on any exam day, across all rooms and both sessions. The database enforces this with a unique index on (version, exam day, teacher) for non-cancelled rows.
H3	Rest days	No teacher on two consecutive exam dates. "Consecutive" means next in the exam-date list, not calendar arithmetic. Default is 1; setting 0 turns the rule off.
H4	Room rotation	No teacher in the same room they had on the immediately preceding exam day.
H5	Duty cap	Default 6 per exam period. Planned and completed duties count toward the cap; cancelled duties do not.
H6	Eligibility	The teacher is active, not on leave that day, not exempt, and not from the department of the room's exam.
H7	Unique slots	One active teacher per room, per session, per staff slot.

> **Why H2 matters for planning:** daily demand is rooms × sessions × staff. For 40 rooms, that is 80 different teachers each day. Back-to-back exam days need twice that (160), because H3 keeps the two days' teachers apart.

3. Allocation

- **Pre-flight check (before generation):** for each exam day, compare duties needed with eligible teachers. Check back-to-back pairs (2 × daily need). Check total capacity (duties needed against eligible teachers × cap). Any shortfall is listed with options: add teachers, make a day a non-exam day, or record an approved exception.
- **Solver:** Google OR-Tools CP-SAT (free, open source), with H1–H7 as hard constraints. The objective is to minimize the highest duty count, then spread the remaining duties evenly.
- **Deterministic:** a seed is stored with each version, so the same inputs and seed produce the same output. Ties are broken by a seeded shuffle, not by teacher ID, so that T001–T080 do not receive every first-day duty.
- **Infeasible schedules:** the system re-solves with H3 and H4 as penalized slack. It reports which rule would need relaxing and by how many teachers. It never relaxes a rule silently; the admin decides, and the exception is logged.
- **Remaining faculty:** every active teacher gets a status for every exam day: Assigned, Available but unassigned, On leave, Exempt, Cap reached, Excluded (department or rest rule), or Standby.

4. Versions and approval

- Generation creates a new version in Draft. Validation runs automatically. The version moves to Approved only after the admin approves it.
- An approved version is never overwritten. Regeneration creates a new version. Edits to an approved version create a revision with a required reason.
- Each version has a counter. An edit made against an old counter is rejected, so two admins cannot overwrite each other's changes.
- The audit log records who made each change, when, and why.

5. Screens

1. **Exam setup:** name, academic year, exam dates, which dates are exam days, sessions with times, staff per room, rest days, and duty cap.
2. **Teachers:** list and search; CSV or Excel import with preview and row-level errors; add, edit, and deactivate; leave dates; exempt flag.
3. **Rooms:** room code, name, exam-department tag, and active flag.
4. **Allotment:** pre-flight panel, Generate, room table filtered by day and session, remaining-faculty list, validation status, Approve, and Print room sheets.
5. **Audit log.**

6. Printed room sheet

One sheet per exam day and session, with: exam name, date, time, room, teacher ID, teacher name, department, a time-in column, and a signature column.

7. Stage 1 acceptance tests

1. Every active room has the configured teachers in every session.
2. No teacher appears twice on the same exam day. This covers morning-plus-afternoon and two rooms in one session.
3. With rest days set to 1, consecutive exam dates never share a teacher.
4. No teacher repeats the same room on the next exam day.
5. No teacher exceeds the cap, and cancelled duties do not count.
6. Inactive, on-leave, exempt, and own-department teachers are never assigned.
7. Pre-flight with 150 teachers and back-to-back Mon + Tue reports a shortfall of 10 and creates no schedule.
8. A manual edit that gives a teacher a second duty on the same day is rejected in the UI, the API, and the database. Each is a separate test.
9. A substitute who already has the other session that day is rejected.
10. The same inputs and seed produce identical output (run twice and compare).
11. Regeneration never changes an approved version.
12. An import with a duplicate ID and a missing department rejects only those rows.

Performance: test at your real size. Optionally, run 200 rooms × 500 teachers once as a stress check to get a measured number.

8. Assumptions until confirmed

- 2 sessions per day, with 1 teacher per room per session.
- Rest days = 1 (no back-to-back exam dates).
- Duty cap = 6 per exam period.
- The own-department rule is on, based on each room's exam-department tag.
- An absent teacher's duty counts toward the cap but not as completed.
- One admin login. Teachers do not log in during Stage 1.
- Stack: Python, FastAPI, SQLite, OR-Tools, and plain HTML/CSS with the approved fonts.

9. What is needed to finalize

1. Confirm or correct the assumptions in section 8.
2. Number of rooms, active teachers, exam days, and session times.
3. The department rule and the list of exemptions.
4. Whether an absent teacher's duty counts toward the cap and whether it is paid.
5. A sample of the room sheet currently in use, if one exists.

Once these are confirmed, Stage 1 starts with the scheduling module and its tests.